

RENITEC
(enalapril maleate/enalaprilat)
[Tablets]

S-LPC-OG0421-MF-102020-Malaysia

LOCAL PHYSICIANS CIRCULAR

Tablets

RENITEC®
Oral (enalapril maleate)

RENITEC® (enalapril maleate) is the maleate salt of enalapril, a derivative of two amino-acids, L-alanine and L-proline. Following oral administration, enalapril is rapidly absorbed and then hydrolyzed to enalaprilat, which is a highly specific, long-acting, non-sulphydryl angiotensin converting enzyme inhibitor.

RENITEC is indicated in the treatment of all grades of essential hypertension, and in renovascular hypertension. It may be used alone as initial therapy or concomitantly with other antihypertensive agents, especially diuretics.

RENITEC is also indicated in the treatment and prevention of heart failure.

A multicenter, placebo-controlled, double-blind study of left ventricular dysfunction (SOLVD) assessed the effects of RENITEC in 6,797 patients. 2,569 patients with all degrees of symptomatic heart failure (primarily mild to moderate New York Heart Association Class II and III) were randomized into the Treatment Arm, and 4,228 patients with asymptomatic left ventricular dysfunction were randomized into the Prevention Arm. The combined results demonstrated an overall reduced risk for the development of major ischemic events. RENITEC decreased the incidence of myocardial infarction and reduced the number of hospitalizations for unstable angina pectoris in patients with left ventricular dysfunction.

In addition, in the Prevention Arm, RENITEC significantly prevented the development of symptomatic heart failure and reduced the number of hospitalizations for heart failure. In the Treatment Arm, RENITEC, as an adjunct to conventional therapy, significantly reduced overall mortality and hospitalization for heart failure and improved NYHA functional class.

In a similar study involving 253 patients with severe heart failure (New York Heart Association Class IV), RENITEC was shown to improve symptoms and reduce mortality significantly.

The cardioprotective properties of RENITEC were demonstrated in these studies by the beneficial effects on survival and retardation of the progression of heart failure in patients with symptomatic heart failure; retardation of the development of symptomatic heart failure in asymptomatic patients with left ventricular dysfunction; and prevention of coronary ischemic events in patients with left ventricular dysfunction, specifically reduction in the incidence of myocardial infarction and reduction in hospitalization for unstable angina pectoris.

PHARMACOKINETICS AND METABOLISM

Oral enalapril is rapidly absorbed, with peak serum concentrations of enalapril occurring within one hour. Based on urinary recovery, the extent of absorption of enalapril from oral enalapril is approximately 60%.

RENITEC
(enalapril maleate/enalaprilat)
[Tablets]

S-LPC-OG0421-MF-102020-Malaysia

Following absorption, oral enalapril is rapidly and extensively hydrolyzed to enalaprilat, a potent angiotensin converting enzyme inhibitor. Similar peak serum concentrations of enalaprilat occur about 4 hours after an oral dose of enalapril. Excretion of enalaprilat is primarily renal. The principal components in urine are enalaprilat, accounting for about 40% of the dose, and intact enalapril. Except for conversion to enalaprilat, there is no evidence for significant metabolism of enalapril. The serum concentration profile of enalaprilat exhibits a prolonged terminal phase, apparently associated with binding to ACE. In subjects with normal renal function, steady state serum concentrations of enalaprilat were achieved by the fourth day of administration of oral enalapril. The effective half-life for accumulation of enalaprilat following multiple doses of oral enalapril is 11 hours. The absorption of oral RENITEC is not influenced by the presence of food in the gastrointestinal tract. The extent of absorption and hydrolysis of enalapril are similar for the various doses in the recommended therapeutic range.

Studies in dogs indicate that enalapril crosses the blood-brain barrier poorly, if at all; enalaprilat does not enter the brain. Multiple doses of oral enalapril in rats do not result in accumulation in any tissues. Milk of lactating rats contains radioactivity following administration of ^{14}C enalapril maleate. Radioactivity was found to cross the placenta following administration of ^{14}C enalapril maleate to pregnant hamsters.

PHARMACODYNAMICS

Administration of RENITEC to patients with hypertension results in a reduction of both supine and standing blood pressure without a significant increase in heart rate.

Symptomatic postural hypotension is infrequent. In some patients the development of optimal blood pressure reduction may require several weeks of therapy. Abrupt withdrawal of RENITEC has not been associated with rapid increase in blood pressure.

Effective inhibition of ACE activity usually occurs 2 to 4 hours after oral administration of an individual dose of enalapril. Onset of antihypertensive activity was usually seen at one hour, with peak reduction of blood pressure achieved by 4 to 6 hours after administration. The duration of effect is dose-related. However, at recommended doses, antihypertensive and hemodynamic effects have been shown to be maintained for at least 24 hours.

The duration of hemodynamic effects appears to be dose-related.

Antihypertensive treatment with RENITEC leads to a significant regression of left ventricular hypertrophy with preservation of left ventricular systolic performance.

In hemodynamic studies in patients with essential hypertension, blood pressure reduction was accompanied by a reduction in peripheral arterial resistance with an increase in cardiac output and little or no change in heart rate. Following administration of RENITEC there was an increase in renal blood flow; glomerular filtration rate was unchanged. There was no evidence of sodium or water retention. However, in patients with low pretreatment glomerular filtration rates, the rates were usually increased.

RENITEC
(enalapril maleate/enalaprilat)
[Tablets]

S-LPC-OG0421-MF-102020-Malaysia

Chronic administration of RENITEC to patients with essential hypertension and renal insufficiency may be associated with improvements in renal function, evidenced by increased glomerular filtration rate.

In short term clinical studies in diabetic and nondiabetic patients with renal disease, decreases in albuminuria and urinary excretion of IgG and total urinary protein were seen after the administration of enalapril.

When given together with thiazide-type diuretics, the blood pressure-lowering effects of RENITEC are at least additive. RENITEC may reduce or prevent the development of thiazide-induced hypokalemia.

Treatment with RENITEC is not usually associated with adverse effects on plasma uric acid.

Treatment with RENITEC has been associated with favorable effects on plasma lipoprotein fractions and favorable or no effect on total cholesterol levels.

In patients with heart failure on therapy with digitalis and diuretics, treatment with oral RENITEC was associated with decreases in peripheral resistance and blood pressure. Cardiac output increased, while heart rate (usually elevated in patients with heart failure) decreased. Pulmonary capillary wedge pressure was also reduced. Exercise tolerance and severity of heart failure, as measured by New York Heart Association criteria, improved. These actions continued during chronic therapy.

In patients with mild to moderate heart failure, enalapril retarded progressive cardiac dilatation/enlargement and failure, as evidenced by reduced left ventricular end diastolic and systolic volumes and improved ejection fraction.

Clinical data have shown that enalapril reduced the frequency of ventricular arrhythmias in patients with heart failure, although the underlying mechanisms and clinical significance are not known.

INDICATIONS

Treatment of:

- * All Grades of Essential Hypertension**
- * Renovascular Hypertension**
- * All Degrees of Heart Failure**

In patients with symptomatic heart failure, RENITEC is also indicated to:

- Improve Survival
 - Retard the Progression of Heart Failure
-

RENITEC
(enalapril maleate/enalaprilat)
[Tablets]

S-LPC-OG0421-MF-102020-Malaysia

- Reduce Hospitalization for Heart Failure

*** Prevention of Symptomatic Heart Failure**

In asymptomatic patients with left ventricular dysfunction, RENITEC is indicated to:

- Retard the Development of Symptomatic Heart Failure
- Reduce Hospitalization for Heart Failure

*** Prevention of Coronary Ischemic Events in Patients with Left Ventricular Dysfunction**

RENITEC is indicated to:

- Reduce the Incidence of Myocardial Infarction
- Reduce Hospitalization for Unstable Angina Pectoris

DOSAGE AND ADMINISTRATION

Oral:

Since absorption of Tablets RENITEC is not affected by food, the tablets may be administered before, during, or after meals.

ESSENTIAL HYPERTENSION

The initial dose is 10 to 20 mg, depending on the degree of hypertension, and is given once daily. In mild hypertension the recommended initial dose is 10 mg daily. For other degrees of hypertension the initial dose is 20 mg daily. The usual maintenance dose is one 20 mg tablet taken once daily. The dosage should be adjusted according to the needs of the patient to a maximum of 40 mg daily.

RENOVASCULAR HYPERTENSION

Since blood pressure and renal function in such patients may be particularly sensitive to ACE inhibition, therapy should be initiated with a lower starting dose (e.g. 5 mg or less). The dosage should then be adjusted according to the needs of the patient. Most patients may be expected to respond to one 20 mg tablet taken once daily. For patients with hypertension who have been treated recently with diuretics, caution is recommended (see next paragraph).

CONCOMITANT DIURETIC THERAPY IN HYPERTENSION

Symptomatic hypotension may occur following the initial dose of RENITEC; this is more likely in patients who are being treated currently with diuretics. Caution is recommended, therefore, since these patients may be volume- or salt-depleted. The diuretic therapy should be discontinued for 2-3 days prior to initiation of therapy with RENITEC. If this is not possible, the initial dose of RENITEC should be low (5 mg or less) to determine the initial effect on the blood pressure. Dosage should then be adjusted according to the needs of the patient.

RENITEC
(enalapril maleate/enalaprilat)
[Tablets]

S-LPC-OG0421-MF-102020-Malaysia

DOSAGE IN RENAL INSUFFICIENCY

Generally, the intervals between the administration of enalapril should be prolonged and/or the dosage reduced.

Renal Status	Creatinine Clearance mL/min	Initial Dose mg/day
Mild Impairment	<80 >30 mL/min.	5 - 10 mg
Moderate Impairment	≤30 >10 mL/min.	2.5 - 5 mg
Severe Impairment. Normally, these patients will be on dialysis.*	≤10 mL/min.	2.5 mg on dialysis days**

* See PRECAUTIONS - Hemodialysis Patients.

** Enalaprilat is dialyzable. Dosage on nondialysis days should be adjusted depending on the blood pressure response.

HEART FAILURE/ASYMPTOMATIC LEFT VENTRICULAR DYSFUNCTION

The initial dose of RENITEC in patients with symptomatic heart failure or asymptomatic left ventricular dysfunction is 2.5 mg, and it should be administered under close medical supervision to determine the initial effect on the blood pressure. RENITEC may be used in the management of symptomatic heart failure usually with diuretics and, where appropriate, digitalis. In the absence of, or after effective management of, symptomatic hypotension following initiation of therapy with RENITEC in heart failure, the dose should be increased gradually to the usual maintenance dose of 20 mg, given in a single dose or two divided doses, as tolerated by the patient. This dose titration may be performed over a 2 to 4 week period, or more rapidly if indicated by the presence of residual signs and symptoms of heart failure. In patients with symptomatic heart failure, this dosage regimen was effective in reducing mortality.

Blood pressure and renal function should be monitored closely both before and after starting treatment with RENITEC (see PRECAUTIONS) because hypotension and (more rarely) consequent renal failure have been reported. In patients treated with diuretics, the dose should be reduced if possible before beginning treatment with RENITEC. The appearance of hypotension after the initial dose of RENITEC does not imply that hypotension will recur during chronic therapy with RENITEC and does not preclude continued use of the drug. Serum potassium also should be monitored (see DRUG INTERACTIONS).

CONTRAINDICATIONS

RENITEC is contraindicated in patients who are hypersensitive to any component of this product and in patients with a history of angioneurotic edema relating to previous treatment with an angiotensin-converting enzyme inhibitor and in patients with hereditary or idiopathic angioedema.

RENITEC
(enalapril maleate/enalaprilat)
[Tablets]

S-LPC-OG0421-MF-102020-Malaysia

RENITEC should not be administered with aliskiren in patients with diabetes (see DRUG INTERACTIONS).

RENITEC is contraindicated in combination with a neprilysin inhibitor (e.g., sacubitril). Do not administer RENITEC within 36 hours of switching to or from sacubitril/valsartan, a product containing a neprilysin inhibitor. (See PRECAUTIONS and DRUG INTERACTIONS.)

PRECAUTIONS

SYMPTOMATIC HYPOTENSION

Symptomatic hypotension was seen rarely in uncomplicated hypertensive patients. In hypertensive patients receiving RENITEC, hypotension is more likely to occur if the patient has been volume - depleted, e.g. by diuretic therapy, dietary salt restriction, dialysis, diarrhea or vomiting (see DRUG INTERACTIONS and SIDE EFFECTS). In patients with heart failure, with or without associated renal insufficiency, symptomatic hypotension has been observed. This is most likely to occur in those patients with more severe degrees of heart failure, as reflected by the use of high doses of loop diuretics, hyponatremia or functional renal impairment. In these patients, therapy should be started under medical supervision and the patients should be followed closely whenever the dose of RENITEC and/or diuretic is adjusted. Similar considerations may apply to patients with ischemic heart or cerebrovascular disease in whom an excessive fall in blood pressure could result in a myocardial infarction or cerebrovascular accident.

If hypotension occurs, the patient should be placed in the supine position and, if necessary, should receive an intravenous infusion of normal saline. A transient hypotensive response is not a contraindication to further doses, which can be given usually without difficulty once the blood pressure has increased after volume expansion.

In some patients with heart failure who have normal or low blood pressure, additional lowering of systemic blood pressure may occur with RENITEC. This effect is anticipated, and usually is not a reason to discontinue treatment. If hypotension becomes symptomatic, a reduction of dose and/or discontinuation of the diuretic and/or RENITEC may be necessary.

AORTIC STENOSIS/HYPERTROPHIC CARDIOMYOPATHY

As with all vasodilators, ACE inhibitors should be given with caution to patients with obstruction in the outflow tract of the left ventricle.

RENAL FUNCTION IMPAIRMENT

In some patients hypotension following the initiation of therapy with ACE inhibitors may lead to some further impairment in renal function. Acute renal failure, usually reversible, has been reported in this situation.

Patients with renal insufficiency may require reduced and/or less frequent doses of RENITEC (see DOSAGE AND ADMINISTRATION). In some patients, with bilateral renal artery stenosis or stenosis of the artery to a solitary kidney, increases of blood urea and serum creatinine, usually

RENITEC
(enalapril maleate/enalaprilat)
[Tablets]

S-LPC-OG0421-MF-102020-Malaysia

reversible upon discontinuation of therapy, have been seen. This is especially likely in patients with renal insufficiency.

Some patients with no apparent pre-existing renal disease have developed usually minor and transient increases in blood urea and serum creatinine when RENITEC has been given concomitantly with a diuretic. Dosage reduction and/or discontinuation of the diuretic and/or RENITEC may be required.

HYPERSENSITIVITY/ANGIONEUROTIC EDEMA

Angioneurotic edema of the face, extremities, lips, tongue, glottis and/or larynx has been reported rarely in patients treated with angiotensin converting enzyme inhibitors, including RENITEC. This may occur at any time during treatment. In such cases, RENITEC should be discontinued promptly and appropriate monitoring should be instituted to ensure complete resolution of symptoms prior to dismissing the patient. Even in those instances where swelling of only the tongue is involved, without respiratory distress, patients may require prolonged observation since treatment with antihistamines and corticosteroids may not be sufficient.

Very rarely, fatalities have been reported due to angioedema associated with laryngeal edema or tongue edema. Patients with involvement of the tongue, glottis or larynx are likely to experience airway obstruction, especially those with a history of airway surgery. Where there is involvement of the tongue, glottis or larynx, likely to cause airway obstruction, appropriate therapy, which may include subcutaneous epinephrine solution 1:1000 (0.3 mL to 0.5 mL) and/or measures to ensure a patent airway, should be administered promptly.

Black patients receiving ACE inhibitors have been reported to have a higher incidence of angioedema compared to non-blacks.

Patients with a history of angioedema unrelated to ACE inhibitor therapy may be at increased risk of angioedema while receiving an ACE inhibitor (also see CONTRAINDICATIONS).

Patients receiving coadministration of ACE inhibitor and mTOR (mammalian target of rapamycin) inhibitor (e.g., temsirolimus, sirolimus, everolimus) therapy may be at increased risk for angioedema.

Patients receiving concomitant ACE inhibitor and neprilysin inhibitor therapy may be at increased risk for angioedema (see CONTRAINDICATIONS and DRUG INTERACTIONS).

Patients receiving concomitant ACE inhibitor and vildagliptin may be at increased risk for angioedema (see DRUG INTERACTIONS).

ANAPHYLACTOID REACTIONS DURING HYMENOPTERA DESENSITIZATION

Rarely, patients receiving ACE inhibitors during desensitization with hymenoptera venom have experienced life-threatening anaphylactoid reactions. These reactions were avoided by temporarily withholding ACE inhibitor therapy prior to each desensitization.

HEMODIALYSIS PATIENTS

RENITEC
(enalapril maleate/enalaprilat)
[Tablets]

S-LPC-OG0421-MF-102020-Malaysia

Anaphylactoid reactions have been reported in patients dialyzed with high-flux membranes (e.g., AN 69⁺) and treated concomitantly with an ACE inhibitor. In these patients consideration should be given to using a different type of dialysis membrane or a different class of antihypertensive agent.

COUGH

Cough has been reported with the use of ACE inhibitors. Characteristically, the cough is nonproductive, persistent and resolves after discontinuation of therapy. ACE inhibitor-induced cough should be considered as part of the differential diagnosis of cough.

SURGERY/ANESTHESIA

In patients undergoing major surgery or during anesthesia with agents that produce hypotension, enalapril blocks angiotensin II formation secondary to compensatory renin release. If hypotension occurs and is considered to be due to this mechanism, it can be corrected by volume expansion.

HYPERKALEMIA – See also DRUG INTERACTIONS, SERUM POTASSIUM.

Risk factors for the development of hyperkalemia include renal insufficiency, diabetes mellitus, and concomitant use of potassium-sparing diuretics (e.g., spironolactone, eplerenone, triamterene, or amiloride), potassium supplements, potassium-containing salt substitutes, or other drugs that may increase serum potassium (e.g., trimethoprim-containing products).

The use of potassium supplements, potassium-sparing diuretics, potassium-containing salt substitutes, or other drugs that may increase serum potassium, particularly in patients with impaired renal function, may lead to a significant increase in serum potassium. Hyperkalemia can cause serious, sometimes fatal, arrhythmias.

If concomitant use of RENITEC and any of the above-mentioned agents is deemed appropriate, they should be used with caution and with frequent monitoring of serum potassium.

HYPOGLYCEMIA

Diabetic patients treated with oral antidiabetic agents or insulin starting an ACE inhibitor should be told to closely monitor for hypoglycemia, especially during the first month of combined use (see DRUG INTERACTIONS).

USE IN PREGNANCY

The use of RENITEC during pregnancy is not recommended. When pregnancy is detected, RENITEC should be discontinued as soon as possible, unless it is considered life-saving for the mother.

In a published retrospective epidemiological study, infants whose mothers had taken an ACE inhibitor drug during the first trimester of pregnancy appeared to have an increased risk of major congenital malformations compared with infants whose mothers had not undergone first trimester exposure to ACE inhibitor drugs. The number of cases of birth defects is small and the findings of this study have not yet been repeated.

RENITEC
(enalapril maleate/enalaprilat)
[Tablets]

S-LPC-OG0421-MF-102020-Malaysia

ACE inhibitors can cause fetal and neonatal morbidity and mortality when administered to pregnant women during the second and third trimesters. Use of ACE inhibitors during this period has been associated with fetal and neonatal injury including hypotension, renal failure, hyperkalemia, and/or skull hypoplasia in the newborn. Maternal oligohydramnios, presumably representing decreased fetal renal function, has occurred and may result in limb contractures, craniofacial deformations and hypoplastic lung development. If RENITEC is used, the patient should be apprised of the potential hazard to the fetus.

These adverse effects to the embryo and fetus do not appear to have resulted from intrauterine ACE-inhibitor exposure limited to the first trimester.

In those rare cases where ACE inhibitor use during pregnancy is deemed essential, serial ultrasound examinations should be performed to assess the intraamniotic environment. If oligohydramnios is detected, RENITEC should be discontinued unless it is considered life-saving for the mother. Patients and physicians should be aware, however, that oligohydramnios may not appear until after the fetus has sustained irreversible injury.

Infants whose mothers have taken RENITEC should be closely observed for hypotension, oliguria and hyperkalemia. Enalapril, which crosses the placenta, has been removed from the neonatal circulation by peritoneal dialysis with some clinical benefit, and theoretically may be removed by exchange transfusion.

NURSING MOTHERS

Enalapril and enalaprilat are secreted in human milk in trace amounts. Caution should be exercised if RENITEC is given to a nursing mother.

PEDIATRIC USE

RENITEC is not indicated for use in children.

DRUG INTERACTIONS

ANTIHYPERTENSIVE THERAPY

Additive effect may occur when RENITEC is used together with other antihypertensive therapy.

SERUM POTASSIUM – See also PRECAUTIONS, HYPERKALEMIA.

In clinical trials, serum potassium usually remained within normal limits. In hypertensive patients treated with RENITEC alone for up to 48 weeks, mean increases in serum potassium of approximately 0.2 mEq/L were observed. In patients treated with RENITEC plus a thiazide diuretic, the potassium-losing effect of the diuretic was attenuated usually by the effect of enalapril.

If RENITEC is given with a potassium-losing diuretic, diuretic-induced hypokalemia may be ameliorated.

RENITEC
(enalapril maleate/enalaprilat)
[Tablets]

S-LPC-OG0421-MF-102020-Malaysia

Risk factors for the development of hyperkalemia include renal insufficiency, diabetes mellitus, and concomitant use of potassium-sparing diuretics (e.g. spironolactone, eplerenone, triamterene or amiloride), potassium supplements, potassium-containing salt substitutes, or other drugs that may increase serum potassium (e.g., trimethoprim-containing products).

The use of potassium supplements, potassium-sparing diuretics, potassium-containing salt substitutes, or other drugs that may increase serum potassium, particularly in patients with impaired renal function, may lead to a significant increase in serum potassium.

If concomitant use of RENITEC and the above-mentioned agents is deemed appropriate, they should be used with caution and with frequent monitoring of serum potassium.

ANTIDIABETICS

Epidemiological studies have suggested that concomitant administration of ACE inhibitors and antidiabetic medicines (insulins, oral hypoglycemic agents) may cause an increased blood-glucose-lowering effect with risk of hypoglycemia. This phenomenon appeared to be more likely to occur during the first weeks of combined treatment and in patients with renal impairment. In diabetic patients treated with oral antidiabetic agents or insulin, glycemic control should be closely monitored for hypoglycemia, especially during the first month of treatment with an ACE inhibitor.

SERUM LITHIUM

As with other drugs which eliminate sodium, lithium clearance may be reduced. Therefore, serum lithium levels should be monitored carefully if lithium salts are to be administered.

NON-STEROIDAL ANTI-INFLAMMATORY DRUGS INCLUDING SELECTIVE CYCLOOXYGENASE-2 INHIBITORS

Non-steroidal anti-inflammatory drugs (NSAIDs) including selective cyclooxygenase-2 inhibitors (COX-2 inhibitors) may reduce the effect of diuretics and other antihypertensive drugs. Therefore, the antihypertensive effect of angiotensin II receptor antagonists or ACE inhibitors may be attenuated by NSAIDs including selective COX-2 inhibitors.

In some patients with compromised renal function (e.g., elderly patients or patients who are volume-depleted including those on diuretic therapy) who are being treated with non-steroidal anti-inflammatory drugs including selective cyclooxygenase-2 inhibitors, the co-administration of angiotensin II receptor antagonists or ACE inhibitors may result in a further deterioration of renal function, including possible acute renal failure. These effects are usually reversible. Therefore, the combination should be administered with caution in patients with compromised renal function.

DUAL BLOCKADE OF THE RENIN-ANGIOTENSIN-ALDOSTERONE SYSTEM

Dual blockade of the renin-angiotensin-aldosterone system (RAAS) with angiotensin receptor blockers, ACE inhibitors, or direct renin inhibitors (such as aliskiren) is associated with increased risks of hypotension, syncope, hyperkalemia, and changes in renal function (including acute renal failure) compared to monotherapy. Closely monitor blood pressure, renal function, and electrolytes in patients on RENITEC and other agents that affect the RAAS. Do not co-

RENITEC
(enalapril maleate/enalaprilat)
[Tablets]

S-LPC-OG0421-MF-102020-Malaysia

administer aliskiren with RENITEC in patients with diabetes. Avoid use of aliskiren with RENITEC in patients with renal impairment (GFR <60ml/min).

GOLD

Nitritoid reactions (symptoms include facial flushing, nausea, vomiting and hypotension) have been reported rarely in patients on therapy with injectable gold (sodium aurothiomalate) and concomitant ACE inhibitor therapy including enalapril.

MAMMALIAN TARGET OF RAPAMYCIN (mTOR) INHIBITORS

Patients taking concomitant mTOR inhibitor (e.g., temsirolimus, sirolimus, everolimus) therapy may be at increased risk for angioedema (see PRECAUTIONS).

NEPRILYSIN INHIBITORS

Patients taking a concomitant neprilysin inhibitor (e.g., sacubitril) may be at increased risk for angioedema (see CONTRAINDICATIONS and PRECAUTIONS).

VILDAGLIPTIN

Patients taking concomitant vildagliptin may be at increased risk for angioedema (see PRECAUTIONS).

SIDE EFFECTS

RENITEC has been demonstrated to be generally well tolerated. In clinical studies, the overall incidence of side effects was no greater with RENITEC than with placebo. For the most part, side effects have been mild and transient in nature, and have not required discontinuation of therapy.

The following side effects have been associated with the use of Tablets RENITEC:

Dizziness and headache were the more commonly reported side effects. Fatigue and asthenia were reported in 2-3% of patients. Other side effects occurred in less than 2% of patients, and included hypotension, orthostatic hypotension, syncope, nausea, diarrhea, muscle cramps, rash, and cough. Less frequently renal dysfunction, renal failure, and oliguria have been reported.

Hypersensitivity/Angioneurotic Edema

Angioneurotic edema of the face, extremities, lips, tongue, glottis and/or larynx has been reported rarely (see PRECAUTIONS). In very rare cases, intestinal angioedema has been reported with angiotensin converting enzyme inhibitors including enalapril.

Side effects which occurred very rarely, either during controlled clinical trials or after the drug was marketed, include:

CARDIOVASCULAR

RENITEC
(enalapril maleate/enalaprilat)
[Tablets]

S-LPC-OG0421-MF-102020-Malaysia

myocardial infarction or cerebrovascular accident, possibly secondary to excessive hypotension in high risk patients (see PRECAUTIONS)

chest pain

palpitations

rhythm disturbances

angina pectoris

Raynaud's phenomenon

ENDOCRINE

syndrome of inappropriate antidiuretic hormone secretion (SIADH)

GASTROINTESTINAL

ileus

pancreatitis

hepatic failure

hepatitis - either hepatocellular or cholestatic

jaundice

abdominal pain

vomiting

dyspepsia

constipation

anorexia

stomatitis

METABOLIC

Cases of hypoglycemia in diabetic patients on oral antidiabetic agents or insulin have been reported (see DRUG INTERACTIONS).

NERVOUS SYSTEM/ PSYCHIATRIC

depression

confusion

somnolence

insomnia

nervousness

paresthesia

vertigo

dream abnormality

RESPIRATORY

pulmonary infiltrates

bronchospasm/asthma

dyspnea

rhinorrhea

sore throat and hoarseness

SKIN

diaphoresis

erythema multiforme

exfoliative dermatitis

Stevens-Johnson syndrome

RENITEC
(enalapril maleate/enalaprilat)
[Tablets]

S-LPC-OG0421-MF-102020-Malaysia

toxic epidermal necrolysis
pemphigus
pruritus
urticaria
alopecia

OTHER

impotence
flushing
taste alteration
tinnitus
glossitis
blurred vision

A symptom complex has been reported which may include some or all of the following: fever, serositis, vasculitis, myalgia/myositis, arthralgia/arthritis, a positive ANA, elevated ESR, eosinophilia, and leukocytosis. Rash, photosensitivity or other dermatologic manifestations may occur.

LABORATORY TEST FINDINGS

Clinically important changes in standard laboratory parameters were rarely associated with administration of RENITEC. Increases in blood urea and serum creatinine, and elevations of liver enzymes and/or serum bilirubin have been seen. These are usually reversible upon discontinuation of RENITEC. Hyperkalemia and hyponatremia have occurred.

Decreases in hemoglobin and hematocrit have been reported.

Since the drug was marketed a small number of cases of neutropenia, thrombocytopenia, bone marrow depression, and agranulocytosis have been reported in which a causal relationship to therapy with RENITEC could not be excluded.

OVERDOSAGE

Limited data are available for overdosage in humans. The most prominent features of overdosage reported to date are marked hypotension, beginning some six hours after ingestion of tablets, concomitant with blockade of the renin-angiotensin system, and stupor. Serum enalaprilat levels 100- and 200-fold higher than usually seen after therapeutic doses have been reported after ingestion of 300 mg and 440 mg of enalapril, respectively.

The recommended treatment of overdosage is intravenous infusion of normal saline solution. If available, angiotensin II infusion may be beneficial. If ingestion is recent, induce emesis. Enalaprilat may be removed from the general circulation by hemodialysis (see PRECAUTIONS, Hemodialysis Patients).

STORAGE

Store Tablets RENITEC below 30°C (86°F).

RENITEC
(enalapril maleate/enalaprilat)
[Tablets]

S-LPC-OG0421-MF-102020-Malaysia

AVAILABILITY

RENITEC (enalapril maleate) 5mg, 10mg and 20mg are available in blister packs of 30's.

SHELF LIFE

Please refer to the expiry date on the outer carton.

APPEARANCE

RENITEC 5mg:	White, barrel shaped tablet, one side scored, the other side RENITEC.
RENITEC 10mg:	Rust red, barrel shaped, tablet, one side scored, the other side RENITEC.
RENITEC 20mg:	Peach, barrel shaped tablet, one side scored, the other side RENITEC.

MANUFACTURED BY

Organon Pharma (UK) Limited
Shotton Lane, Cramlington,
Northumberland, NE23 3JU,
United Kingdom.

PRODUCT REGISTRATION HOLDER

ORGANON MALAYSIA SDN. BHD.
3A-08-02, Level 8, Corporate Tower 3A,
Pavilion Damansara Heights,
No. 3, Jalan Damanlela
50490 Kuala Lumpur, Malaysia.

DATE OF REVISION: 15 October 2025

Copyright 2025 Organon group of companies. All rights reserved.
